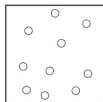
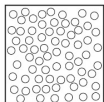
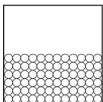


3.3.2 How structure affects properties

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 3			
The three states of matter are solid, liquid and gas. Melting and freezing take place at the melting point, boiling and condensing take place at the boiling point. The three states of matter can be represented by a simple model. In this model, particles are represented by small solid spheres.  Solid Liquid Gas	Students should be able to use models of particles as small spheres to represent the three states of matter.	5.2.2.1	4.1.1.1

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
When a solid melts to become a liquid the particles are able to move about but stay close together. When a liquid boils and becomes a gas the particles separate and move about rapidly. Substances with high melting points have strong forces that hold their particles together. Substances with low boiling points have weak forces between their particles.	Suggested activity for TDA Compare the melting points of a range different substances, eg candle wax, beeswax polish, butter, margarine, cooking fat.	5.2.2.1	4.6.2.5
Outcome 4			
Diamond and graphite are forms of the element carbon with different properties because of their different structures. Diamond is hard because the carbon atoms are joined together in a giant three-dimensional structure. Graphite is slippery because the carbon atoms are joined together in layers that can slide over each other.	Students should be able to recognise diamond and graphite from diagrams of their structures.	5.2.3.1 and 5.2.3.2	4.8.1.1